

Ransomware

Case Study: Ransomware Attack on a College Student System

Ransomware is a type of cybercrime in which an attacker blocks access to a system or data and demands payment (ransom) to restore it. A realistic scenario involves a college student downloading a “free” software tool from an untrusted website.

The attack typically happens in the following steps:

First, the user downloads a file that appears legitimate but contains hidden malware. Second, once installed, the ransomware silently runs in the background and starts encrypting files such as documents, images, and projects. Third, after encryption is complete, a message appears on the screen stating that the files are locked and can only be recovered by paying a ransom, usually in cryptocurrency. Finally, the attacker may threaten to delete or leak data if payment is not made within a limited time.

Students, small businesses, and individuals with limited cybersecurity awareness are common targets. They are often chosen because they may not have strong security systems or regular data backups.

The consequences can be serious. Victims may lose important academic work, personal files, or sensitive data. Financial loss is also possible if the ransom is paid, with no guarantee of recovery. Additionally, such attacks can cause stress, loss of productivity, and potential data privacy risks.

This case highlights the importance of safe internet practices, such as avoiding untrusted downloads, using antivirus software, and regularly backing up data.